

Chapter 8

Large Deviations for Sparse Graphs

The development in this monograph till now has been based on results from graph limit theory. This theory, however, is inadequate for understanding the behavior of sparse graphs. The goal of this chapter is to describe an alternative approach, called nonlinear large deviations, that allows us to prove similar results for sparse graphs. Nonlinear large deviation theory gives a way of getting quantitative error bounds in some of the large deviation theorems proved in earlier chapters. The quantitative error bounds make it possible to extend the results to the sparse regime. At the time of writing this monograph, this theory is not as well-developed as the theory for dense graphs, but developed enough to make some progress about sparse graphs.

8.1 An Approximation for Partition Functions

Take any $N \geq 1$. Let $\|f\|$ denote the supremum norm of any function $f : [0, 1]^N \rightarrow \mathbb{R}$. Suppose that $f : [0, 1]^N \rightarrow \mathbb{R}$ is twice continuously differentiable in $(0, 1)^N$, such that f and all its first and second order derivatives extend continuously to the boundary. For each i and j , let

$$f_i := \frac{\partial f}{\partial x_i} \quad \text{and} \quad f_{ij} := \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

Define

$$a := \|f\|, \quad b_i := \|f_i\| \quad \text{and} \quad c_{ij} := \|f_{ij}\|.$$

Given $\epsilon > 0$, let $\mathcal{D}(\epsilon)$ be a finite subset of $\mathbb{R}^N$ such that for all $x \in \{0, 1\}^N$, there exists $d = (d_1, \dots, d_N) \in \mathcal{D}(\epsilon)$ such that

$$\sum_{i=1}^N (f_i(x) - d_i)^2 \leq N\epsilon^2. \tag{8.1.1}$$

Define

$$F := \log \sum_{x \in \{0,1\}^N} e^{f(x)}.$$

In the terminology of statistical mechanics, F is the logarithm of the partition function of the probability measure on $\{0, 1\}^N$ with Hamiltonian f . For $x \in [0, 1]$, let

$$I(x) := x \log x + (1 - x) \log(1 - x), \quad (8.1.2)$$

and for $x = (x_1, \dots, x_N) \in [0, 1]^N$, let

$$I(x) := \sum_{i=1}^N I(x_i).$$

The following theorem gives a sufficient condition on f under which the approximation

$$F = \sup_{x \in [0,1]^N} (f(x) - I(x)) + \text{lower order terms} \quad (8.1.3)$$

is valid. This is sometimes called the ‘naive mean field approximation’ in the statistical physics literature. Roughly speaking, the condition is that in addition to some smoothness assumptions, the gradient vector $\nabla f(x) = (\partial f / \partial x_1, \dots, \partial f / \partial x_N)$ may be approximately encoded by $o(N)$ bits of information. A bit more precisely, we need $|\mathcal{D}(\epsilon)| = e^{o(N)}$ for some $\epsilon = o(1)$, where the implicit assumption is that $N \rightarrow \infty$ and f varies with N . We will refer to this as the ‘low complexity gradient’ condition.

Theorem 8.1 *Let f , a , b_i , c_{ij} , $\mathcal{D}(\epsilon)$, F and I be as above. Then for any $\epsilon > 0$,*

$$F \leq \sup_{x \in [0,1]^N} (f(x) - I(x)) + \text{complexity} + \text{smoothness},$$

where

$$\begin{aligned} \text{complexity} &= \frac{1}{4} \left(N \sum_{i=1}^N b_i^2 \right)^{1/2} \epsilon + 3N\epsilon + \log |\mathcal{D}(\epsilon)|, \text{ and} \\ \text{smoothness} &= 4 \left(\sum_{i=1}^N (ac_{ii} + b_i^2) + \frac{1}{4} \sum_{i,j=1}^N (ac_{ij}^2 + b_i b_j c_{ij} + 4b_i c_{ij}) \right)^{1/2} \\ &\quad + \frac{1}{4} \left(\sum_{i=1}^N b_i^2 \right)^{1/2} \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2} + 3 \sum_{i=1}^N c_{ii} + \log 2. \end{aligned}$$

Moreover, F satisfies the lower bound

$$F \geq \sup_{x \in [0,1]^N} (f(x) - I(x)) - \frac{1}{2} \sum_{i=1}^N c_{ii}.$$

The rest of this section is devoted to the proof of Theorem 8.1. Examples are worked in subsequent sections.

We will generally denote the i th coordinate of a vector $x \in \mathbb{R}^N$ by x_i . Similarly, the i th coordinate of a random vector X will be denoted by X_i . Given $x \in [0, 1]^N$, define $x^{(i)}$ to be the vector $(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_N)$. For a random vector X define $X^{(i)}$ similarly. Given a function $g : [0, 1]^N \rightarrow \mathbb{R}$, define the discrete derivative $\Delta_i g$ as

$$\begin{aligned} \Delta_i g(x) &:= g(x_1, \dots, x_{i-1}, 1, x_{i+1}, \dots, x_N) \\ &\quad - g(x_1, \dots, x_{i-1}, 0, x_{i+1}, \dots, x_N). \end{aligned} \quad (8.1.4)$$

For each i , define a function $\hat{x}_i : [0, 1]^N \rightarrow [0, 1]$ as

$$\hat{x}_i(x) = \frac{1}{1 + e^{-\Delta_i f(x)}}.$$

Let $\hat{x} : [0, 1]^N \rightarrow [0, 1]^N$ be the vector-valued function whose i th coordinate function is $\hat{x}_i$. When the vector x is understood from the context, we will simply write $\hat{x}$ and $\hat{x}_i$ instead of $\hat{x}(x)$ and $\hat{x}_i(x)$. The proof of Theorem 8.1 requires two key lemmas.

Lemma 8.1 *Let $X = (X_1, \dots, X_N)$ be a random vector that has probability density proportional to $e^{f(x)}$ on $\{0, 1\}^N$ with respect to the counting measure. Let $\hat{X} = \hat{x}(X)$. Then*

$$\mathbb{E}[(f(X) - f(\hat{X}))^2] \leq \sum_{i=1}^N (ac_{ii} + b_i^2) + \frac{1}{4} \sum_{i,j=1}^N (ac_{ij}^2 + b_i b_j c_{ij}).$$

Proof It is easy to see that

$$\hat{x}_i(x) = \mathbb{E}(X_i \mid X_j = x_j, 1 \leq j \leq N, j \neq i).$$

Let $D := f(X) - f(\hat{X})$. Then clearly

$$|D| \leq 2a. \quad (8.1.5)$$

Define

$$h(x) := f(x) - f(\hat{x}(x)),$$

so that $D = h(X)$. Note that for $i \neq j$,

$$\frac{\partial \hat{x}_j}{\partial x_i} = \frac{e^{-\Delta_j f(x)}}{(1 + e^{-\Delta_j f(x)})^2} \int_0^1 f_{ij}(x_1, \dots, x_{j-1}, t, x_{j+1}, \dots, x_n) dt,$$

and for $i = j$, the above derivative is identically equal to zero. Since

$$\frac{e^{-x}}{(1 + e^{-x})^2} \leq \frac{1}{4}$$

for all $x \in \mathbb{R}$, this shows that for all i and j ,

$$\left\| \frac{\partial \hat{x}_j}{\partial x_i} \right\| \leq \frac{c_{ij}}{4}. \quad (8.1.6)$$

Thus,

$$\begin{aligned} \left\| \frac{\partial h}{\partial x_i} \right\| &\leq \|f_i\| + \sum_{j=1}^N \|f_j\| \left\| \frac{\partial \hat{x}_j}{\partial x_i} \right\| \\ &\leq b_i + \frac{1}{4} \sum_{j=1}^N b_j c_{ij}. \end{aligned} \quad (8.1.7)$$

Consequently, if $D_i := h(X^{(i)})$, then

$$|D - D_i| \leq b_i + \frac{1}{4} \sum_{j=1}^N b_j c_{ij}. \quad (8.1.8)$$

For $t \in [0, 1]$ and $x \in [0, 1]^N$ define

$$u_i(t, x) := f_i(tx + (1 - t)\hat{x}),$$

so that

$$h(x) = \int_0^1 \sum_{i=1}^N (x_i - \hat{x}_i) u_i(t, x) dt.$$

Thus,

$$\mathbb{E}(D^2) = \int_0^1 \sum_{i=1}^N \mathbb{E}((X_i - \hat{X}_i) u_i(t, X) D) dt. \quad (8.1.9)$$

Now,

$$\|u_i\| \leq b_i, \quad (8.1.10)$$

and by (8.1.6),

$$\begin{aligned} \left\| \frac{\partial u_i}{\partial x_i} \right\| &\leq t \|f_{ii}\| + (1-t) \sum_{j=1}^N \|f_{ij}\| \left\| \frac{\partial \hat{x}_j}{\partial x_i} \right\| \\ &\leq t c_{ii} + \frac{1-t}{4} \sum_{j=1}^N c_{ij}^2. \end{aligned} \quad (8.1.11)$$

The bounds (8.1.5), (8.1.8), (8.1.10) and (8.1.11) imply that

$$\begin{aligned} &|\mathbb{E}((X_i - \hat{X}_i)u_i(t, X)D) - \mathbb{E}((X_i - \hat{X}_i)u_i(t, X^{(i)})D_i)| \\ &\leq \mathbb{E}|(u_i(t, X) - u_i(t, X^{(i)}))D| + \mathbb{E}|u_i(t, X^{(i)})(D - D_i)| \\ &\leq 2atc_{ii} + \frac{a(1-t)}{2} \sum_{j=1}^N c_{ij}^2 + b_i^2 + \frac{1}{4} \sum_{j=1}^N b_i b_j c_{ij}. \end{aligned}$$

But $u_i(t, X^{(i)})D_i$ is a function of the random variables $(X_j)_{j \neq i}$ only. Therefore by the definition of $\hat{X}_i$,

$$\mathbb{E}((X_i - \hat{X}_i)u_i(t, X^{(i)})D_i) = 0.$$

Thus,

$$|\mathbb{E}((X_i - \hat{X}_i)u_i(t, X)D)| \leq 2atc_{ii} + \frac{a(1-t)}{2} \sum_{j=1}^N c_{ij}^2 + b_i^2 + \frac{1}{4} \sum_{j=1}^N b_i b_j c_{ij}.$$

Using this bound in (8.1.9) gives

$$\begin{aligned} \mathbb{E}(D^2) &\leq \int_0^1 \sum_{i=1}^N \left(2atc_{ii} + \frac{a(1-t)}{2} \sum_{j=1}^N c_{ij}^2 + b_i^2 + \frac{1}{4} \sum_{j=1}^N b_i b_j c_{ij} \right) dt \\ &= \sum_{i=1}^N (ac_{ii} + b_i^2) + \frac{1}{4} \sum_{i,j=1}^N (ac_{ij}^2 + b_i b_j c_{ij}), \end{aligned}$$

completing the proof. $\square$

Lemma 8.2 *Let all notation be as in Lemma 8.1. Then*

$$\mathbb{E} \left[\left(\sum_{i=1}^N (X_i - \hat{X}_i) \Delta_i f(X) \right)^2 \right] \leq \sum_{i=1}^N b_i^2 + \frac{1}{4} \sum_{i,j=1}^N b_i(b_j + 4)c_{ij}.$$

Proof Let g_i denote the function $\Delta_i f$, for notational simplicity. Note that

$$g_i(x) = \int_0^1 f_i(x_1, \dots, x_{i-1}, t, x_{i+1}, \dots, x_N) dt,$$

which shows that

$$\|g_i\| \leq \|f_i\| = b_i \quad (8.1.12)$$

and for all j ,

$$\left\| \frac{\partial g_i}{\partial x_j} \right\| \leq \|f_{ij}\| = c_{ij}. \quad (8.1.13)$$

Let

$$G(x) := \sum_{i=1}^N (x_i - \hat{x}_i(x)) g_i(x).$$

Then

$$\frac{\partial G}{\partial x_i} = \sum_{j=1}^N \left[\left(1_{\{j=i\}} - \frac{\partial \hat{x}_j}{\partial x_i} \right) g_j(x) + (x_j - \hat{x}_j) \frac{\partial g_j}{\partial x_i} \right]$$

and therefore by (8.1.6), (8.1.12) and (8.1.13),

$$\left\| \frac{\partial G}{\partial x_i} \right\| \leq b_i + \frac{1}{4} \sum_{j=1}^N c_{ij} b_j + \sum_{j=1}^N c_{ij}. \quad (8.1.14)$$

Note that for any x ,

$$|G(x) - G(x^{(i)})| \leq \left\| \frac{\partial G}{\partial x_i} \right\|. \quad (8.1.15)$$

Again, $g_i(X)$ and $G(X^{(i)})$ are both functions of $(X_j)_{j \neq i}$ only. Therefore

$$\mathbb{E}((X_i - \hat{X}_i) g_i(X) G(X^{(i)})) = 0. \quad (8.1.16)$$

Combining (8.1.14), (8.1.15) and (8.1.16) gives

$$\begin{aligned}\mathbb{E}(G(X)^2) &= \sum_{i=1}^N \mathbb{E}((X_i - \hat{X}_i)g_i(X)G(X)) \\ &\leq \sum_{i=1}^N b_i \left(b_i + \frac{1}{4} \sum_{j=1}^N c_{ij}b_j + \sum_{j=1}^N c_{ij} \right).\end{aligned}$$

This completes the proof of the lemma. $\square$

With the aid of Lemma 8.1 and 8.2, we are now ready to prove Theorem 8.1.

Proof (Proof of the Upper Bound in Theorem 8.1) For $x, y \in [0, 1]^N$, let

$$g(x, y) := \sum_{i=1}^N (x_i \log y_i + (1 - x_i) \log(1 - y_i)). \quad (8.1.17)$$

Note that

$$g(x, \hat{x}) - I(\hat{x}) = \sum_{i=1}^N (x_i - \hat{x}_i) \log \frac{\hat{x}_i}{1 - \hat{x}_i} = \sum_{i=1}^N (x_i - \hat{x}_i) \Delta_i f(x). \quad (8.1.18)$$

Let

$$B := 4 \left(\sum_{i=1}^N (ac_{ii} + b_i^2) + \frac{1}{4} \sum_{i,j=1}^N (ac_{ij}^2 + b_i b_j c_{ij} + 4b_i c_{ij}) \right)^{1/2}.$$

Let

$$A_1 := \{x \in \{0, 1\}^N : |I(\hat{x}) - g(x, \hat{x})| \leq B/2\},$$

and

$$A_2 := \{x \in \{0, 1\}^N : |f(x) - f(\hat{x})| \leq B/2\}.$$

Let $A = A_1 \cap A_2$. By Lemma 8.1 and the identity (8.1.18), $\mathbb{P}(X \notin A_1) \leq 1/4$. By Lemma 8.2, $\mathbb{P}(X \notin A_2) \leq 1/4$. Thus, $\mathbb{P}(X \in A) \geq 1/2$, which is the same as

$$\frac{\sum_{x \in A} e^{f(x)}}{\sum_{x \in \{0, 1\}^N} e^{f(x)}} \geq \frac{1}{2}.$$

Therefore by the definition of the set A ,

$$\begin{aligned} F &= \log \sum_{x \in \{0,1\}^N} e^{f(x)} \leq \log \sum_{x \in A} e^{f(x)} + \log 2 \\ &\leq B + \log \sum_{x \in A} e^{f(\hat{x}) - I(\hat{x}) + g(x, \hat{x})} + \log 2. \end{aligned} \quad (8.1.19)$$

Now take some $x \in \{0, 1\}^N$ and let d satisfy (8.1.1). Then by the Cauchy–Schwarz inequality,

$$\sum_{i=1}^N |f_i(x) - d_i| \leq N\epsilon.$$

Fix such an x and d . Note that for each i ,

$$\begin{aligned} |\Delta_i f(x) - f_i(x)| &\leq \int_0^1 |f_i(x_1, \dots, x_{i-1}, t, x_{i+1}, \dots, x_N) - f_i(x)| dt \\ &\leq \|f_{ii}\| = c_{ii}. \end{aligned}$$

By the last two inequalities and (8.1.1),

$$\sum_{i=1}^N |\Delta_i f(x) - d_i| \leq N\epsilon + \sum_{i=1}^N c_{ii}. \quad (8.1.20)$$

and

$$\left(\sum_{i=1}^N (\Delta_i f(x) - d_i)^2 \right)^{1/2} \leq N^{1/2}\epsilon + \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2}. \quad (8.1.21)$$

Let $u(x) = 1/(1 + e^{-x})$. Note that for all x ,

$$|u'(x)| = \frac{1}{(e^{x/2} + e^{-x/2})^2} \leq \frac{1}{4}.$$

Therefore if a vector $p = p(d)$ is defined as $p_i = u(d_i)$, then by (8.1.21),

$$\begin{aligned} \left(\sum_{i=1}^N (\hat{x}_i - p_i)^2 \right)^{1/2} &\leq \left(\frac{1}{16} \sum_{i=1}^N (\Delta_i f(x) - d_i)^2 \right)^{1/2} \\ &\leq \frac{N^{1/2}\epsilon}{4} + \frac{1}{4} \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2}. \end{aligned}$$

Thus, if

$$L := \left(\sum_{i=1}^N b_i^2 \right)^{1/2},$$

then

$$\begin{aligned} |f(\hat{x}) - f(p)| &\leq L \left(\sum_{i=1}^N (\hat{x}_i - p_i)^2 \right)^{1/2} \\ &\leq \frac{LN^{1/2}\epsilon}{4} + \frac{L}{4} \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2}. \end{aligned} \quad (8.1.22)$$

Next, let $v(x) = \log(1 + e^{-x})$. Then for all x ,

$$|v'(x)| = \frac{e^{-x}}{1 + e^{-x}} \leq 1.$$

Consequently,

$$|\log \hat{x}_i - \log p_i| \leq |\Delta_i f(x) - d_i|$$

and

$$|\log(1 - \hat{x}_i) - \log(1 - p_i)| \leq |\Delta_i f(x) - d_i|.$$

Therefore by (8.1.20),

$$|g(x, \hat{x}) - g(x, p)| \leq 2 \sum_{i=1}^N |\Delta_i f(x) - d_i| \leq 2N\epsilon + 2 \sum_{i=1}^N c_{ii}. \quad (8.1.23)$$

Finally, let $w(x) = I(u(x))$. Then

$$\begin{aligned} w'(x) &= u'(x)I'(u(x)) \\ &= \frac{e^{-x}}{(1 + e^{-x})^2} \log \frac{u(x)}{1 - u(x)} \\ &= \frac{xe^{-x}}{(1 + e^{-x})^2}. \end{aligned}$$

Thus, for all x ,

$$|w'(x)| \leq \sup_{x \in \mathbb{R}} \frac{|x|e^{-x}}{(1 + e^{-x})^2} \leq \sup_{x \geq 0} xe^{-x} = \frac{1}{e}.$$

Consequently,

$$|I(\hat{x}_i) - I(p_i)| \leq \frac{1}{e} |\Delta_i f(x) - d_i|,$$

and so by (8.1.20),

$$|I(\hat{x}) - I(p)| \leq \frac{N\epsilon}{e} + \frac{1}{e} \sum_{i=1}^N c_{ii}. \quad (8.1.24)$$

For each $d \in \mathcal{D}(\epsilon)$ let $\mathcal{C}(d)$ be the set of all $x \in \{0, 1\}^N$ such that (8.1.1) holds, and let $p(d)$ be the vector p defined above. Then by (8.1.22), (8.1.23) and (8.1.24),

$$\begin{aligned} \log \sum_{x \in A} e^{f(\hat{x}) - I(\hat{x}) + g(x, \hat{x})} &\leq \log \sum_{d \in \mathcal{D}(\epsilon)} \sum_{x \in \mathcal{C}(d)} e^{f(\hat{x}) - I(\hat{x}) + g(x, \hat{x})} \\ &\leq \frac{LN^{1/2}\epsilon}{4} + \frac{L}{4} \left(\sum_{i=1}^N c_{ii}^2 \right)^{1/2} + 2N\epsilon + 2 \sum_{i=1}^N c_{ii} + \frac{N\epsilon}{e} + \frac{1}{e} \sum_{i=1}^N c_{ii} \\ &\quad + \log \sum_{d \in \mathcal{D}(\epsilon)} \sum_{x \in \mathcal{C}(d)} e^{f(p(d)) - I(p(d)) + g(x, p(d))}. \end{aligned} \quad (8.1.25)$$

Now note that for any $p \in [0, 1]^N$,

$$\sum_{x \in \{0, 1\}^N} e^{g(x, p)} = 1.$$

Thus,

$$\begin{aligned} \log \sum_{d \in \mathcal{D}(\epsilon)} \sum_{x \in \mathcal{C}(d)} e^{f(p(d)) - I(p(d)) + g(x, p(d))} \\ \leq \log \sum_{d \in \mathcal{D}(\epsilon)} e^{f(p(d)) - I(p(d))} \\ \leq \log |\mathcal{D}(\epsilon)| + \sup_{p \in [0, 1]^N} (f(p) - I(p)). \end{aligned} \quad (8.1.26)$$

Combining (8.1.19), (8.1.25) and (8.1.26), the proof is complete. $\square$

Proof (Proof of the Lower Bound in Theorem 8.1) Let y be a point in the cube $[0, 1]^N$. Let $Y = (Y_1, \dots, Y_N)$ be a random vector with independent components, where Y_i is a Bernoulli(y_i) random variable. Then by Jensen's inequality,

$$\begin{aligned}
\sum_{x \in \{0,1\}^N} e^{f(x)} &= \sum_{x \in \{0,1\}^N} e^{f(x) - g(x,y) + g(x,y)} \\
&= \mathbb{E}(e^{f(Y) - g(Y,y)}) \\
&\geq \exp(\mathbb{E}(f(Y) - g(Y,y))) \\
&= \exp(\mathbb{E}(f(Y)) - I(y)).
\end{aligned}$$

Let $S := f(Y) - f(y)$. For $t \in [0, 1]$ and $x \in [0, 1]^N$ define

$$v_i(t, x) := f_i(tx + (1-t)y),$$

so that

$$S = \int_0^1 \sum_{i=1}^N (Y_i - y_i) v_i(t, Y) dt. \quad (8.1.27)$$

By the independence of Y_i and $Y^{(i)}$,

$$\begin{aligned}
|\mathbb{E}((Y_i - y_i) v_i(t, Y))| &= |\mathbb{E}((Y_i - y_i)(v_i(t, Y) - v_i(t, Y^{(i)})))| \\
&\leq \left\| \frac{\partial v_i}{\partial x_i} \right\| \leq t c_{ii}.
\end{aligned}$$

Using this bound in (8.1.27) gives

$$\mathbb{E}(S) \geq - \int_0^1 \sum_{i=1}^N t c_{ii} dt = - \frac{1}{2} \sum_{i=1}^N c_{ii}.$$

This completes the proof. $\square$

8.2 Gradient Complexity of Homomorphism Densities

The goal of this section is to compute estimates for the smoothness and complexity terms of Theorem 8.1 for homomorphism densities. These estimates will be used later for establishing quantitative versions of some of the theorems proved earlier in this monograph, which, in turn, will be used for understanding large deviations in sparse graphs.

Let $n \geq 2$ be a positive integer. Let $\mathcal{P}_n$ denote the set of all upper triangular arrays like $x = (x_{ij})_{1 \leq i < j \leq n}$, where each $x_{ij} \in [0, 1]$. Let H be a finite simple graph with vertex set $V = \{1, 2, \dots, k\}$ and edge set E . For $x \in \mathcal{P}_n$, let

$$t(H, x) := \frac{1}{n^k} \sum_{q_1, \dots, q_k=1}^n \prod_{\{l, l'\} \in E} x_{q_l q_{l'}}, \quad (8.2.1)$$